Assignment Machine Learning 2026

FinGuard Analytics and the Consumer Complaints Challenge

Context

On a cold Monday morning in January, the executive committee at Meridian Financial Group gathered for an emergency strategy session. The mood was tense. Over the past eighteen months, customer complaints had surged across multiple product lines—credit cards, personal loans, digital banking, and mortgage servicing.

At first, leadership believed the increase was a temporary side effect of digital transformation. The company had launched new mobile services, migrated customer support operations, and expanded its online onboarding channels. But the trend did not reverse. Complaints continued to grow, regulators began requesting additional reports, and customer sentiment scores declined.

What concerned executives most was not the volume of complaints—it was the uncertainty. No one could confidently explain why some complaints escalated into serious disputes while others were resolved quickly.

The Chief Risk Officer argued that credit card products were responsible for most unresolved cases, citing anecdotal evidence from internal reports. The Head of Digital Experience insisted that complaints submitted online tended to escalate more frequently than those handled through traditional channels. Meanwhile, the Compliance Director warned that response times varied significantly across states, raising the risk of regulatory scrutiny and potential penalties.

The CEO listened quietly before asking a simple question:

“Which of these assumptions is actually supported by data?”

Silence followed.

Despite the organization’s investment in analytics, complaint handling remained largely reactive. Teams focused on resolving individual cases, but no system existed to anticipate escalation or identify patterns across products, geographies, and customer segments. Decisions were often based on intuition, internal anecdotes, or fragmented reports.

The board recognized that this approach was no longer sustainable. Complaint management had evolved from an operational issue into a strategic priority, touching customer experience, compliance, and brand reputation. A single unresolved complaint could now trigger regulatory investigations, media attention, and customer attrition.

To address the problem, Meridian partnered with FinGuard Analytics, a consultancy specializing in data-driven decision making for financial institutions. FinGuard was given access to the Consumer Financial complaint database—a large, real-world repository capturing thousands of customer complaints across the financial services industry. The dataset included product categories, issue types, submission channels, geographic information, company responses, and resolution indicators.

FinGuard assembled a cross-functional taskforce composed of data scientists, business analysts, and risk specialists. Their mandate was clear: transform complaint data into actionable intelligence. The organization needed to understand what drives complaint escalation, identify patterns hidden in operational data, and build predictive tools capable of anticipating which cases were most likely to result in escalation (disputed).

The stakes were high.

If the taskforce succeeded, Meridian could shift from reactive complaint handling to proactive intervention—prioritizing high-risk cases, improving customer outcomes, and reducing regulatory exposure. If it failed, the organization risked continued reputational damage, operational inefficiencies, and increased compliance costs.

You are a member of the FinGuard Data Science Taskforce. Your role is to analyze the complaint dataset, uncover meaningful patterns, build predictive models, and provide insights that will shape how financial institutions manage customer issues going forward.

Your work will directly influence executive decisions, operational processes, and long-term strategy.

Instructions

1. **Download the Dataset**: Use the complains data set from Moodle
2. **Complete All Tasks**: Follow the tasks outlined below step-by-step.
3. **Prepare Files for Submission**: Submit your work as a ZIP file named: `StudentNumber_Complaints_Assignment.zip`. This ZIP file should contain:
    - A Jupyter Notebook file (`StudentNumber_Complaints_Notebook.ipynb`).
    - A trained model saved as a Pickle file (`StudentNumber_Pipeline.pkl`).
4. **Upload to Moodle**: Submit the ZIP file on Moodle under the appropriate assignment link before the deadline.

Your notebook must clearly demonstrate how you explored the data, prepared features, built models, evaluated performance, and interpreted results.

Question 1 [*4 points*]

Senior executives at Meridian Financial Group have expressed concerns about the drivers of complaint escalation. They suspect that certain financial products generate more negative outcomes, that complaints submitted through digital channels escalate more frequently, and that regional differences may influence response practices.

Q1.1 Using the dataset provided, investigate at least one of these assumptions through exploratory analysis. Determine whether the claim is supported by evidence and quantify your findings [*1 point*].

Q1.2 In addition, identify three insights emerging from your analysis that could inform managerial decision making [*3 points*]. Your response must include both code and interpretation.

Question 2 [*4 points*]

Meridian wants to move beyond reactive complaint management toward a predictive approach that identifies high-risk cases early.

Q2.1 Begin by identifying the type of machine learning problem represented in this context [*1 point*].

Q2.2 Then develop two predictive models capable of estimating whether a complaint will result in an unfavorable outcome. Your analysis must include preprocessing steps, feature engineering, data splitting, and hyperparameter tuning. You must also justify your modeling choices from both a technical and business perspective. [*3 points*]

The executive team needs a robust predictive solution to identify customers most at risk of disputing response. Machine learning models can provide the insights required for targeted retention strategies.

Question 3 [*4 points*]

Q3.1 Executives need to understand whether the predictive models are reliable enough for operational use. Select the evaluation metric or metrics you consider most appropriate and justify your choice in relation to business risk, regulatory implications, and decision making [*2 points*].

Q3.2 Compare the performance of the models you developed and determine which one is best suited for deployment. Explain the practical implications of your results. [*2 points*]

Question 4. [*4 points*]

The selected model will be integrated into Meridian's complaint management system. To prepare for deployment, you must save your model pipeline as a Pickle file and ensure it can be reloaded in a new environment. You must also provide a requirements file listing dependencies and a Python script containing the feature engineering function.

After saving the model, reload it and apply it to an external validation dataset with the same structure as the original data. The model must produce predictions in binary format, where zero represents non-escalated complaints and one represents escalated complaints.

Models will be evaluated based on functionality, reproducibility, and performance on the external dataset.

The model should predict complain using binary outputs: `0` (Not disputed) and `1` (Disputed).

- Save the best-performing model as a Pickle file (`StudentNumber_Pipeline.pkl`). Please see the code `Assignment1_SaveAsPickle.ipynb` with the examples
- To ensure the code runs in every machine, please save the requirements files with the name `StudentNumber_requirements.txt` – please see `Assignment1_SaveAsPickle.ipynb` on how to do it.
  - For this to happen you will also require submitting a .py file with your feature engineer function, please keep the following name `feature_engineering.py`
- If you are using a model outside the scikit-learn you will need to save the following files:
  - `StudentNumber_Preprocessor.pkl` – where the preprocessing steps are orchestrated
  - `feature_engineering.py` – where you should include the feature transformations
  - `StudentNumber_Model.pkl` – where the model does the prediction task.
- Write code to reload the saved model and test it on the external validation dataset (please see `Assignment1_SaveAsPickle.ipynb` on how to do it). Note: loading and testing must be performed in a new notebook.
  - The validation set will be in the format from the file `complaints_modeltesting100.csv`). If your model runs in this dataset it will run in the final validation set.
- Ensure the model outputs predictions in the required format `0` (Not Disputed) and `1` (Disputed).

How This Question Will Be Evaluated:

1. **Top 25% Models**: The models that perform the best on the external validation dataset will receive **full marks for this task**. Performance will be ranked based on F1 score.

   Deleted: accuracy
2. **Next 25% Models**: Students whose models perform in the following 25% will have 3 and so on.
3. **Non-Functional Models**: If your model fails to load, run, or provide predictions in the required format (0 for Not Disputed, 1 for Disputed), you will receive **zero marks for this task**. Make sure your .pkl file and code are functional, tested, and well-documented.
   1. **You can find a test set in the assignment on moodle with the same format as the final** (`complaints_modeltesting100.csv`).

Question 5. [*4 points*]

Meridian's leadership team expects insights that go beyond predictive performance. Based on your analysis and modeling results, identify the factors that most strongly influence complaint escalation [*2 points*].

Using these findings, propose one strategic action that financial institutions could implement to reduce escalation risk, improve customer experience, and strengthen regulatory compliance. Your recommendations should be realistic, evidence-based, and clearly connected to your analysis. [*2 points*]

Annex: data dictionary

Field name	Description	Data type	Notes
Date received	The date the CFPB received the complaint	date & time	
Product	The type of product the consumer identified in the complaint	plain text	This field is a categorical variable.
Sub-product	The type of sub-product the consumer identified in the complaint	plain text	This field is a categorical variable. Not all Products have Sub-products.
Issue	The issue the consumer identified in the complaint	plain text	This field is a categorical variable. Possible values are dependent on Product.
Sub-issue	The sub-issue the consumer identified in the complaint	plain text	This field is a categorical variable. Possible values are dependent on product and issue. Not all Issues have

Field name	Description	Data type	Notes
			corresponding Sub-issues.
Consumer complaint narrative	Consumer complaint narrative is the consumer-submitted description of "what happened" from the complaint. Consumers must opt-in to share their narrative. We will not publish the narrative unless the consumer consents, and consumers can opt-out at any time. The CFPB takes reasonable steps to scrub personal information from each complaint that could be used to identify the consumer.	plain text	Consumers' descriptions of what happened are included if consumers consent to publishing the description and after we take steps to remove personal information.
Company public response	The company's optional, public-facing response to a consumer's complaint. Companies can choose to select a response from a pre-set list of options that will be posted on the	plain text	Companies' public-facing responses to complaints are included if companies choose to publish one. Companies may select a public response from a set list of options as soon as they respond to the complaint, but no

Field name	Description	Data type	Notes
	public database. For example, "Company believes complaint is the result of an isolated error."		later than 180 days after the complaint was sent to the company for response.
Company	The complaint is about this company	plain text	This field is a categorical variable.
State	The state of the mailing address provided by the consumer	plain text	This field is a categorical variable.
ZIP code	The mailing ZIP code provided by the consumer	plain text	The mailing ZIP code provided by the consumer. The 5-digit United States Postal Service ZIP code will be published where provided unless the consumer lived in a ZIP code aligned to a United States Census Bureau ZIP Code Tabulation Area (ZCTA) with fewer than 20,000 people and consented to publication of their complaint narrative. In those cases, where the Census ZCTA had fewer than 20,000 people, the 3-digit ZIP code will be published

Field name	Description	Data type	Notes
			if the 3-digit ZCTA has more than 20,000 people. Otherwise, no ZIP code will be published.
Tags	Data that supports easier searching and sorting of complaints submitted by or on behalf of consumers. For example, complaints where the submitter reports the age of the consumer as 62 years or older are tagged, ‘Older American.’ Complaints submitted by or on behalf of a servicemember or the spouse or dependent of a servicemember are tagged, ‘Servicemember.’ Servicemember includes anyone who is active duty, National Guard, or Reservist, as well as anyone who previously served	plain text	

Field name	Description	Data type	Notes
	and is a Veteran or retiree.		
Consumer consent provided?	Identifies whether the consumer opted in to publish their complaint narrative. We do not publish the narrative unless the consumer consents and consumers can opt-out at any time.	plain text	This field shows whether a consumer provided consent to publish their complaint narrative, as listed below: **Consent provided:** Consumer opted in to share their complaint narrative. Data populates in this field 60 days after the complaint was sent to the company for response or after the company provides an optional company public response – whichever comes first, and after steps have been taken to scrub personal information from the complaint narrative. **Consent not provided:** Consumer did not opt-in to publish their complaint narrative. Data populates in this field

Field name	Description	Data type	Notes
			60 days after the complaint was sent to the company for response or after the company provides an optional company public response – whichever comes first. **Consent withdrawn:** Consumer opted in to publish their complaint narrative and later withdrew their consent. **N/A:** Consumers did not have the option to publish their consumer complaint narrative or the complaint was received before March 19, 2015. Data populates in this field immediately. **Other:** Complaint does not meet criteria for narrative publication. Blanks appear until at least 60 days after the complaint is sent to the company for response or until the company provides an optional

Field name	Description	Data type	Notes
			company public response – whichever comes first.
Submitted via	How the complaint was submitted to the CFPB	plain text	This field is a categorical variable.
Date sent to company	The date the CFPB sent the complaint to the company	date & time	
Company response to consumer	This is how the company responded. For example, "Closed with explanation."	plain text	This field is a categorical variable.
Timely response?	Whether the company gave a timely response	plain text	yes/no
Consumer disputed?	Whether the consumer disputed the company's response	plain text	**Yes** **No** **N/A:** The Bureau discontinued the consumer dispute option on April 24, 2017.
Complaint ID	The unique identification number for a complaint	number	

Note: dataset based on https://www.consumerfinance.gov/complaint/